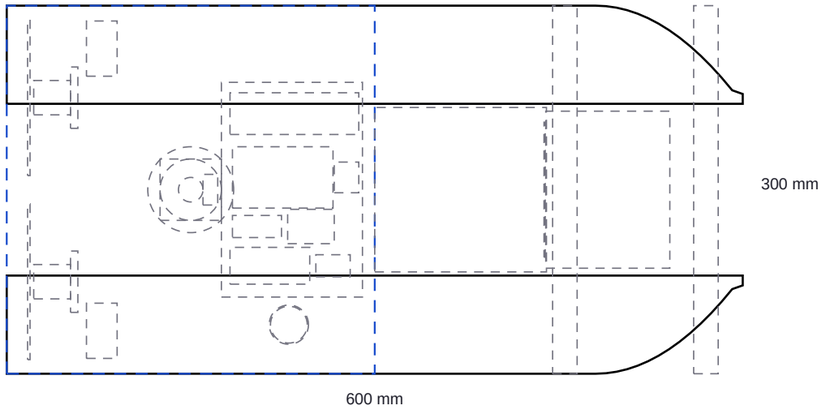
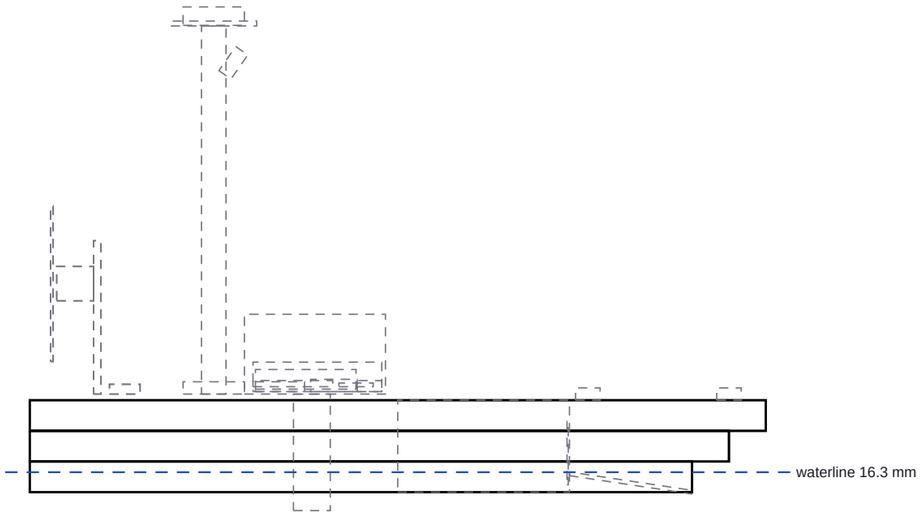
VARUNA Twin-Hull Skimmer

600 x 300 mm thermocol boat | Recommended

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 396 mm to GPS top
Weight	1289 g empty, 1558 g with 269 g of plastic
Draft	16.3 mm empty, 19.7 mm full
Reserve buoyancy	4.7 kg before the deck goes under
Stability (GM)	671 mm: very stable
Trim	bow 14.9 mm higher than stern empty; slide the battery to level
Collection	1.34 L basket
Camera	334 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	1 sheet 1000 x 500 x 25 mm

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

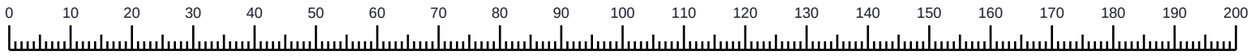
cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers



ruler in mm: the 0 to 200 marks must be exactly 200 mm apart

100 mm x 100 mm

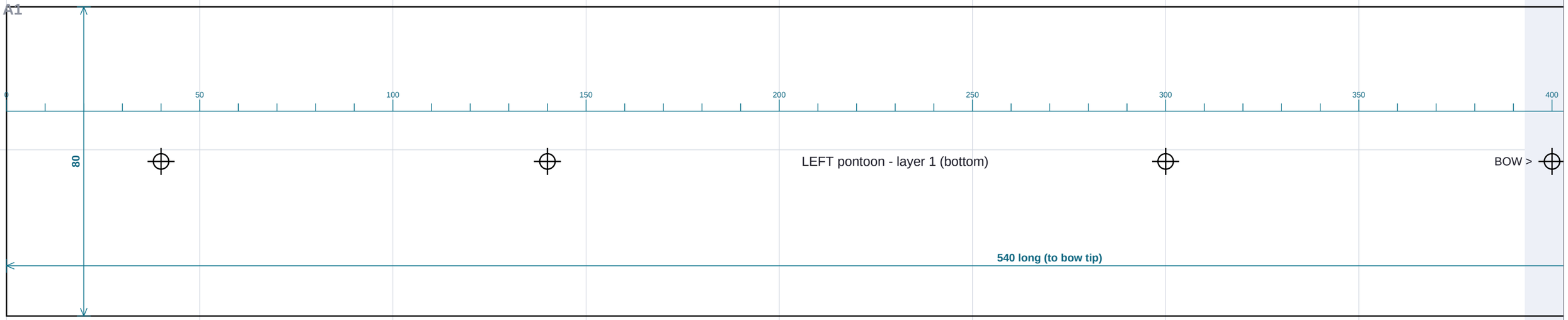
How to print on A3 so every part comes out real size

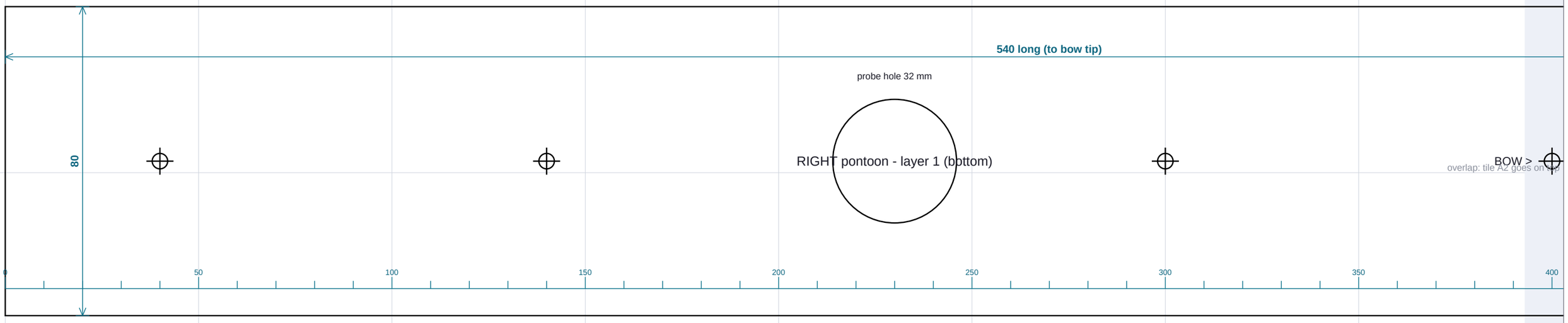
1. Paper: A3, landscape. Printer: 'A3' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do not cut

A1


0
50
100
150
200
250
300
350
400
80
LEFT pontoon - layer 1 (bottom)
BOW >
540 long (to bow tip)


540 long (to bow tip)
probe hole 32 mm
80
RIGHT pontoon - layer 1 (bottom)
BOW >
overlap: tile A2 goes on top
0
50
100
150
200
250
300
350
400

<- must measure 50 mm

A2

400 450 500

ramp wedge
ramp wedge

400 450 500

A1

0 50 100 150 200 250 300 350 400

80

LEFT pontoon - layer 2 (middle)

570 long (to bow tip)

570 long (to bow tip)

probe hole 32 mm

80

RIGHT pontoon - layer 2 (middle)

overlap: tile A2 goes on top

0 50 100 150 200 250 300 350 400

A2

400 450 500

BOW >

BOW >

400 450 500

<- must measure 50 mm

A1

0 50 100 150 200 250 300 350 400

80

LEFT pontoon - layer 3 (top)

600 long (to bow tip)

basket lip rests here
basket lip rests here

600 long (to bow tip)

probe hole 32 mm

80

RIGHT pontoon - layer 3 (top)

overlap: tile A2 goes on top

0 50 100 150 200 250 300 350 400

A2

400 450 500 550

BOW >

BOW >

400 450 500 550

<- must measure 50 mm

A1

300 long

ESC 20 A

Motor tower

STERN

Mast socket

Waterproof electronics box

300 wide

BOW >

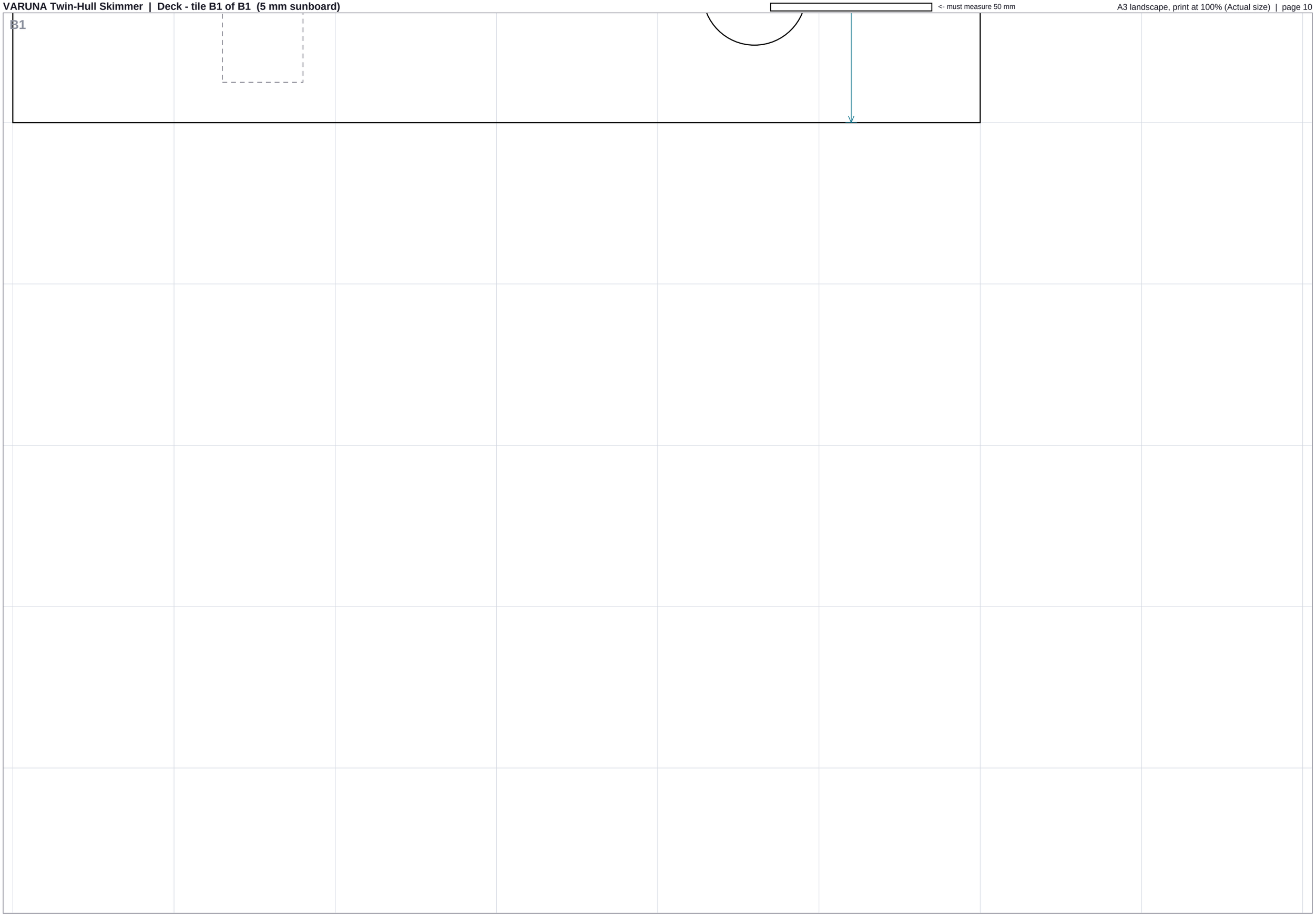
0 50 100 150 200 250 300

Motor tower

probe hole

ESC 20 A

overlap: tile B1 goes on top

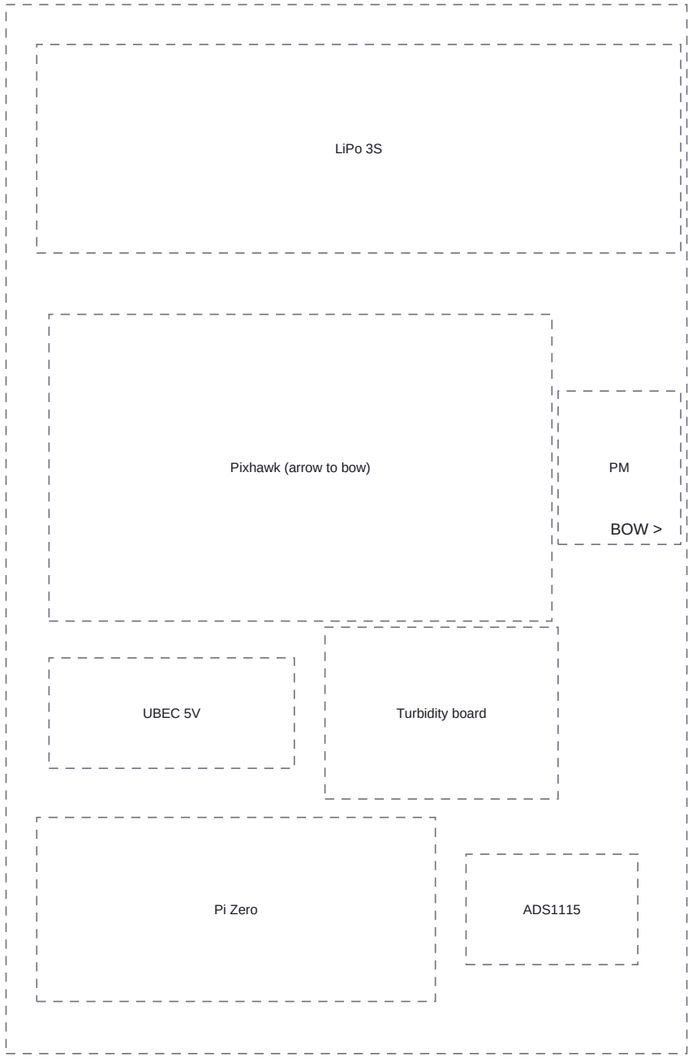

B1
<- must measure 50 mm

VARUNA Twin-Hull Skimmer | small parts 1 of 4

<- must measure 50 mm

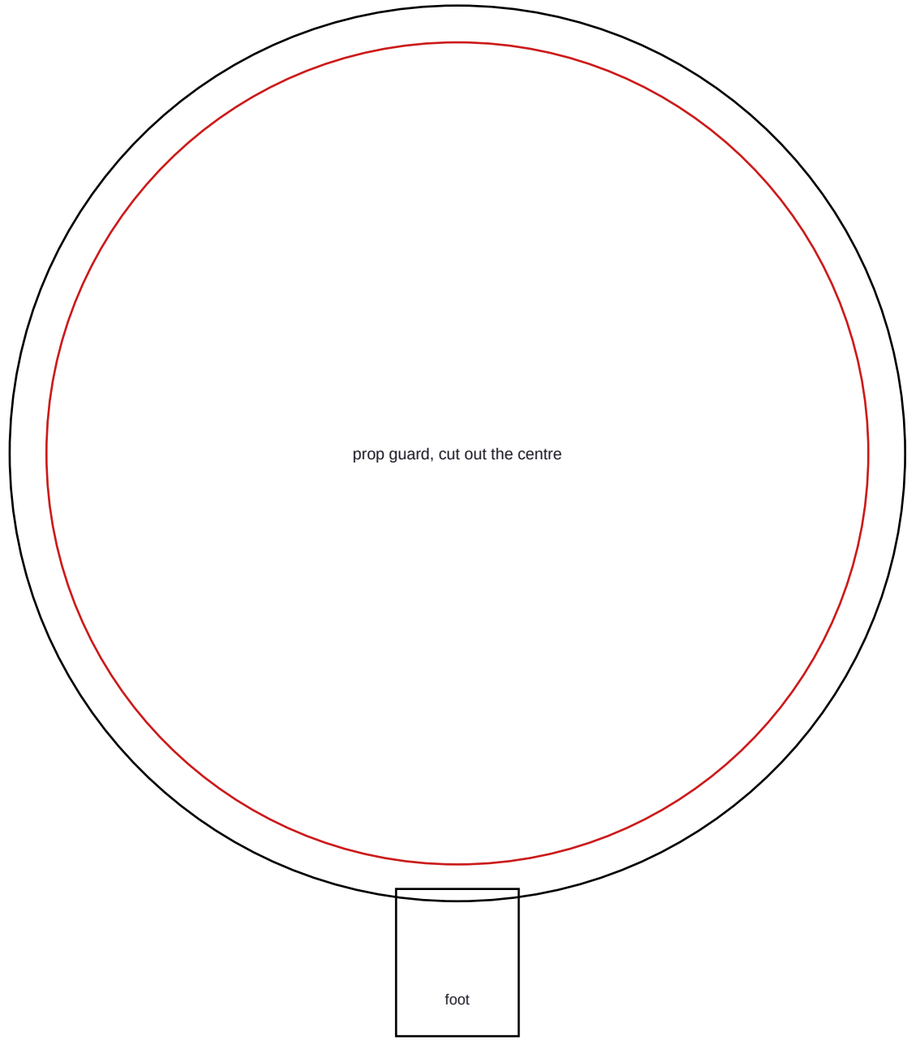
Electronics box floor guide

paper guide, glue inside the floor | 111 x 171 mm



Prop guard ring (left)

5 mm sunboard | 146 x 168 mm

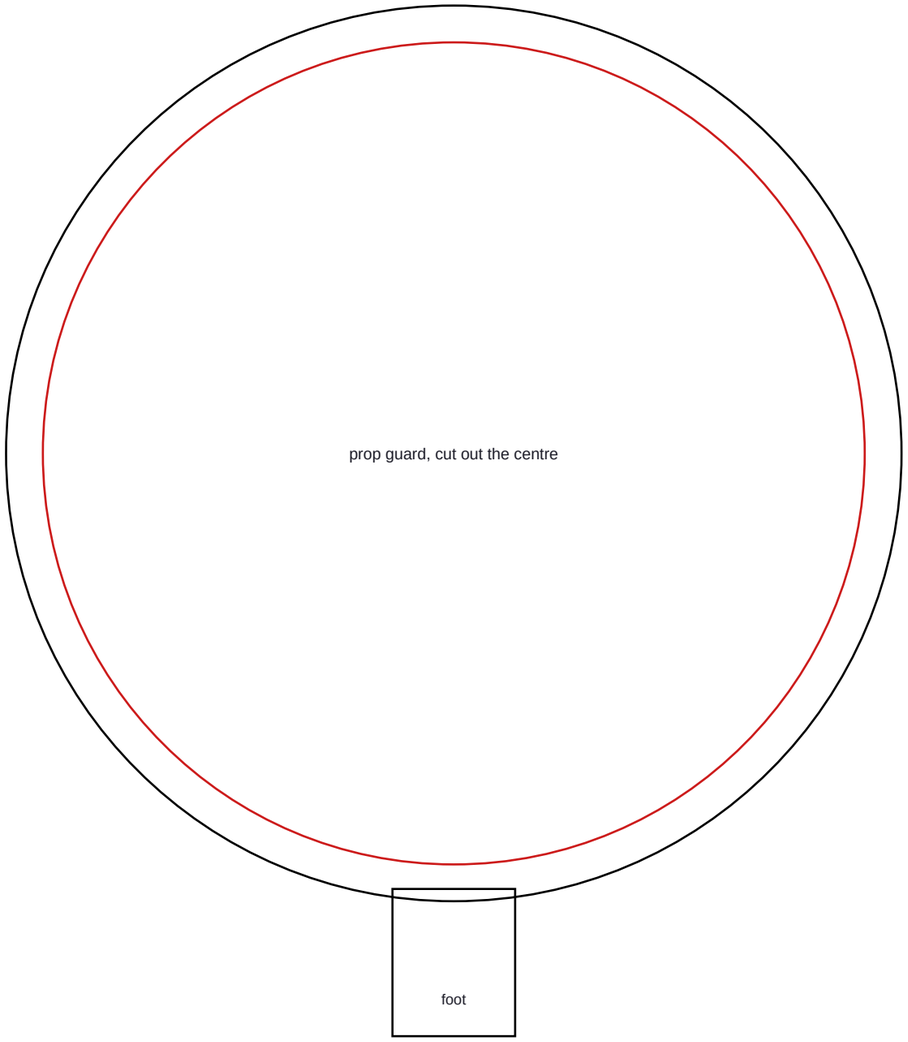


VARUNA Twin-Hull Skimmer | small parts 2 of 4

<- must measure 50 mm

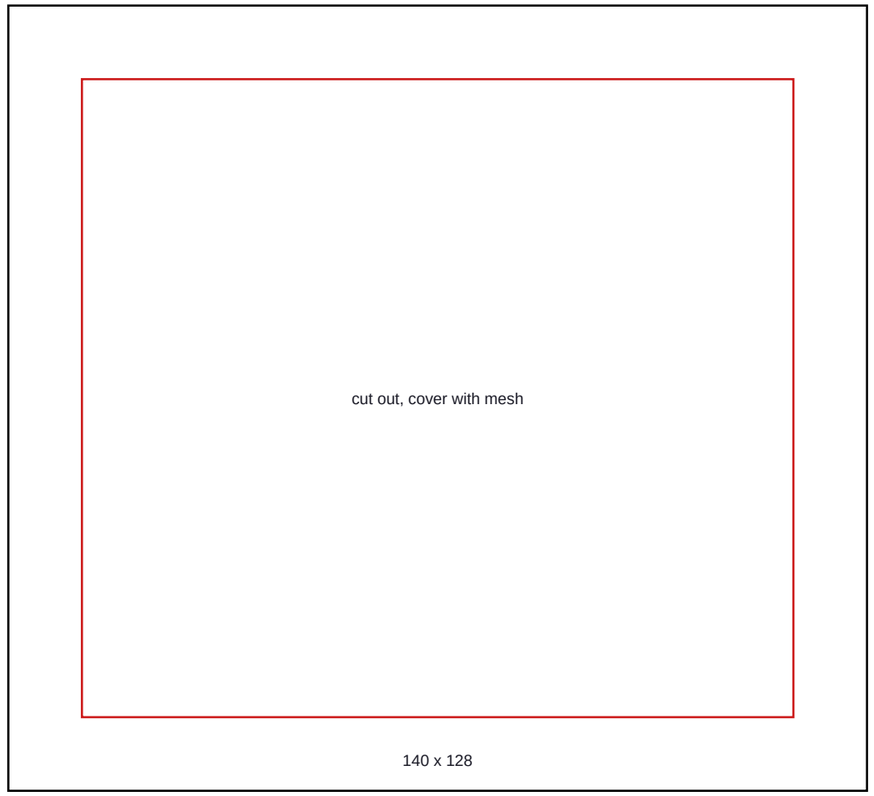
Prop guard ring (right)

5 mm sunboard | 146 x 168 mm



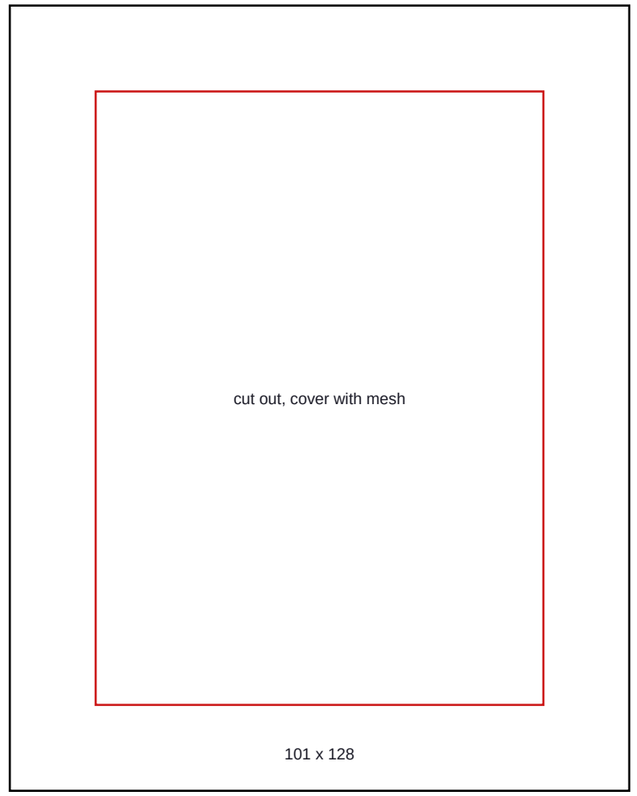
Basket bottom

3 mm sunboard | 140 x 128 mm



Intake ramp frame

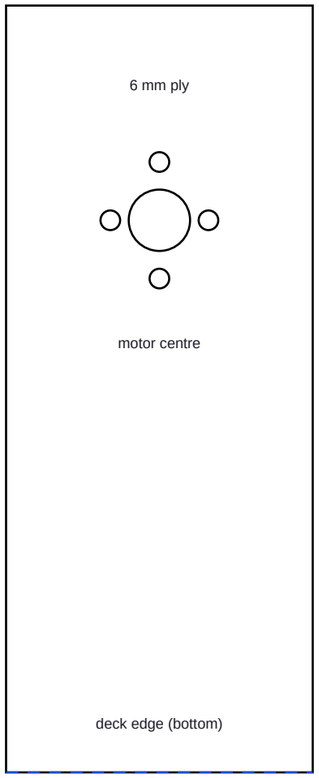
3 mm sunboard | 101 x 128 mm



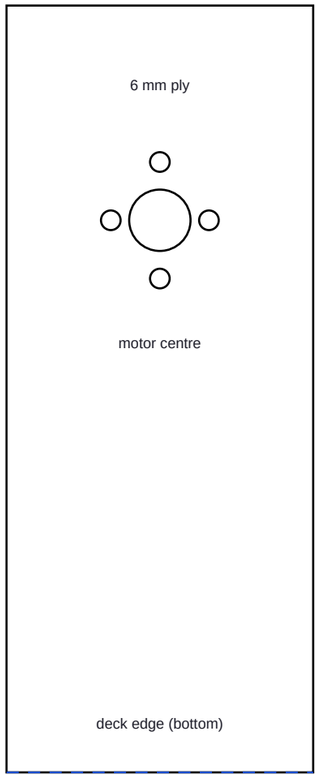
VARUNA Twin-Hull Skimmer | small parts 3 of 4

<- must measure 50 mm

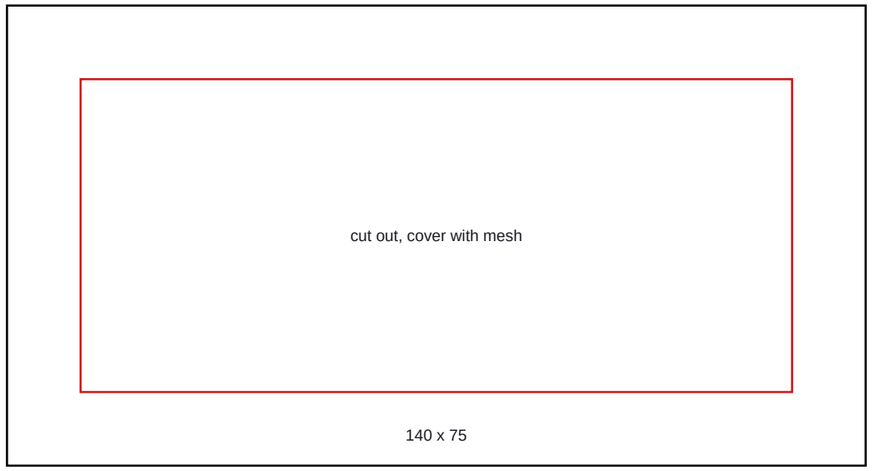
Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



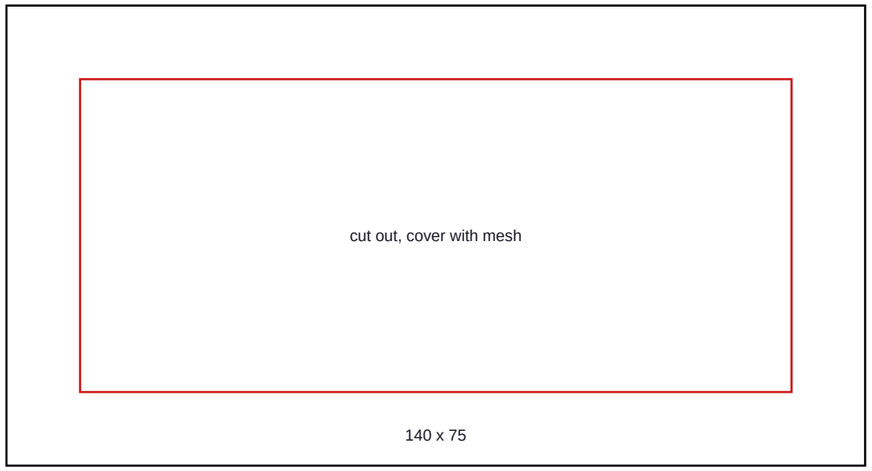
Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



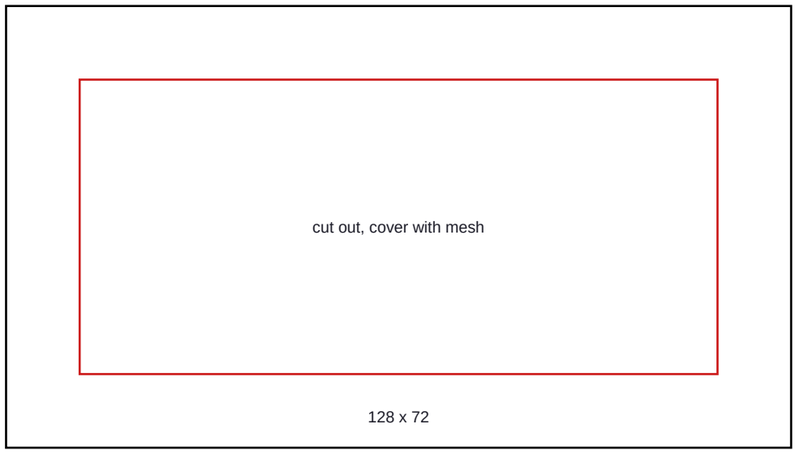
Basket side (left)
3 mm sunboard | 140 x 75 mm



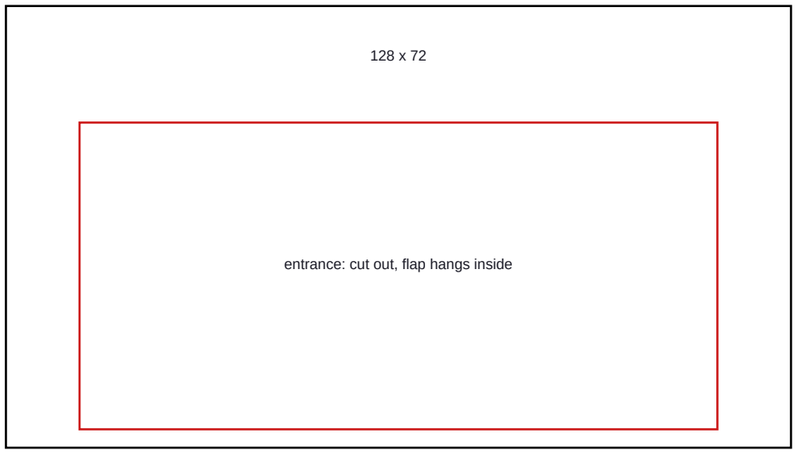
Basket side (right)
3 mm sunboard | 140 x 75 mm



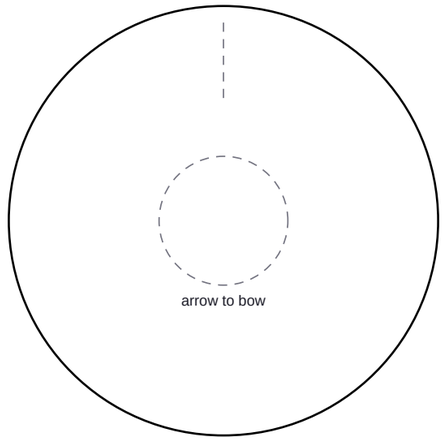
Basket back end
3 mm sunboard | 128 x 72 mm



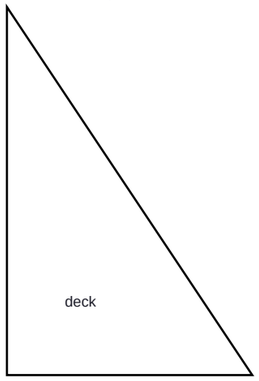
Basket front end (entrance)
3 mm sunboard | 128 x 72 mm



GPS disc
5 mm sunboard | 70 x 70 mm



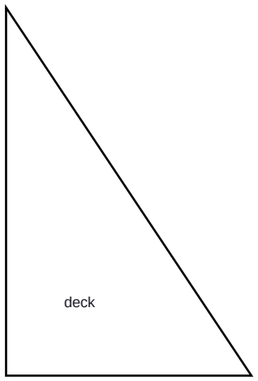
Tower gusset
5 mm sunboard | 40 x 60 mm



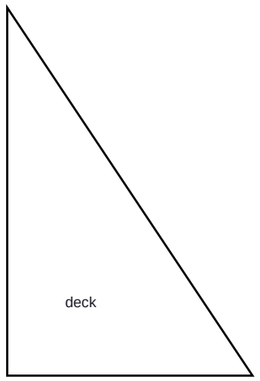
VARUNA Twin-Hull Skimmer | small parts 4 of 4

<- must measure 50 mm

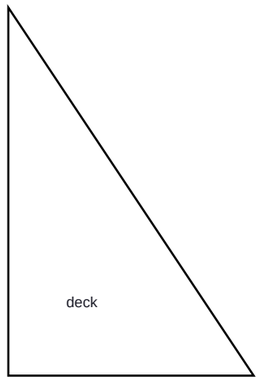
Tower gusset
5 mm sunboard | 40 x 60 mm



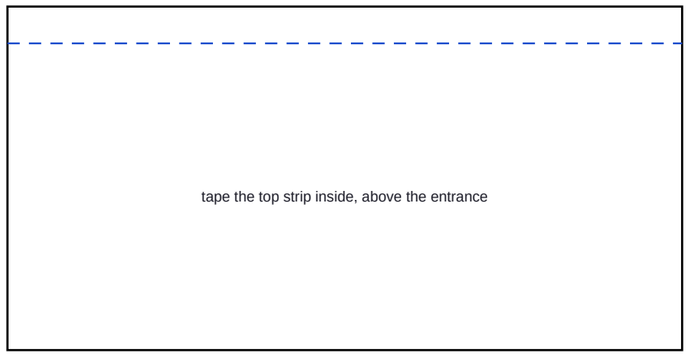
Tower gusset
5 mm sunboard | 40 x 60 mm



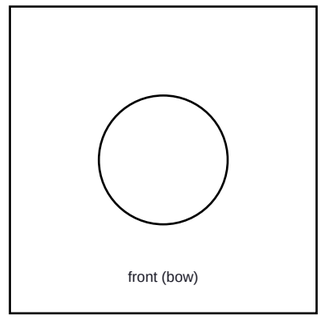
Tower gusset
5 mm sunboard | 40 x 60 mm



One-way flap
PET sheet from a bottle / transparency | 110 x 56 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
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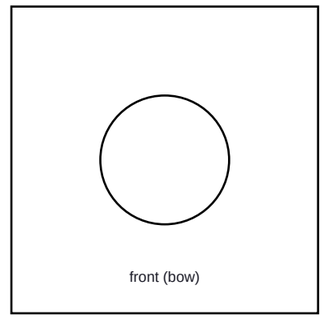
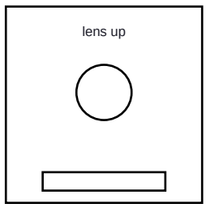


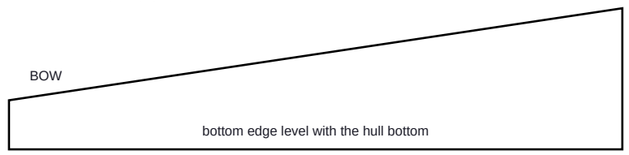
Camera plate
3 mm sunboard | 32 x 32 mm



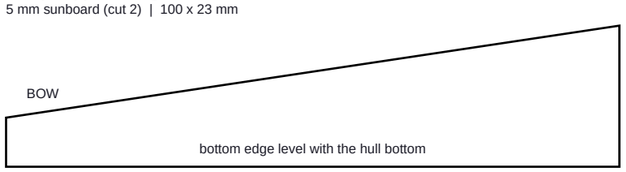
Basket lip (left)
3 mm sunboard | 140 x 28 mm

Basket lip (right)
3 mm sunboard | 140 x 28 mm

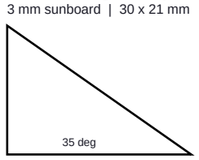
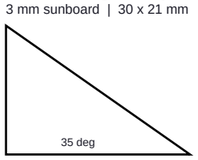
Ramp support wedge
5 mm sunboard (cut 2) | 100 x 23 mm



Ramp support wedge
5 mm sunboard (cut 2) | 100 x 23 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm



Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm